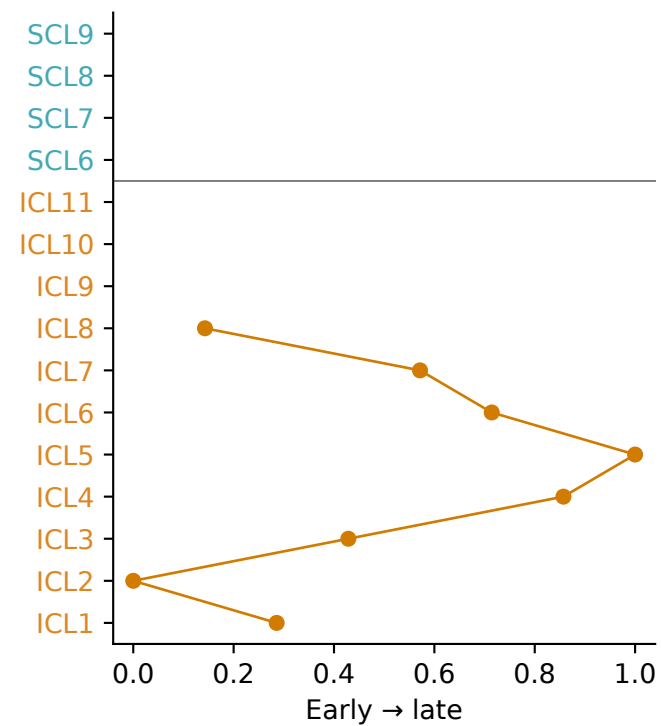
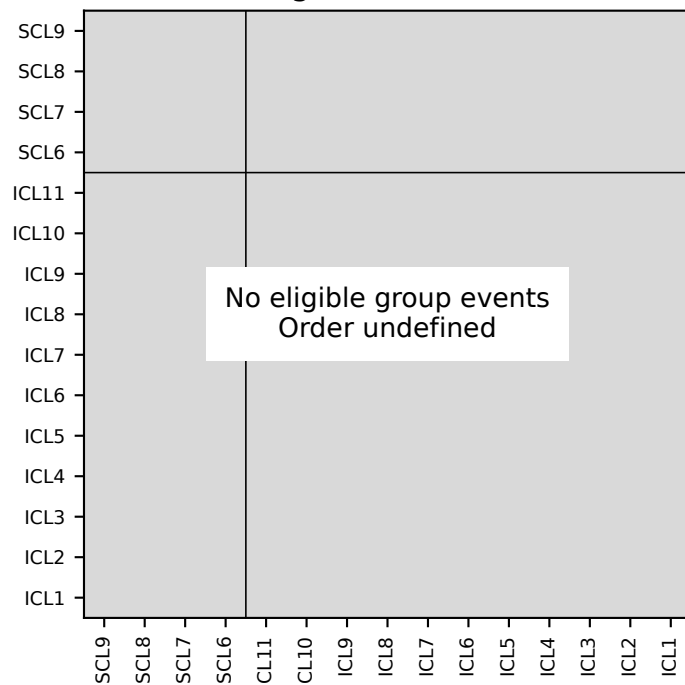


Frozen contact observer on two same-J periodic solutions
One solution per branch; eight repeated cycles, not independent trials

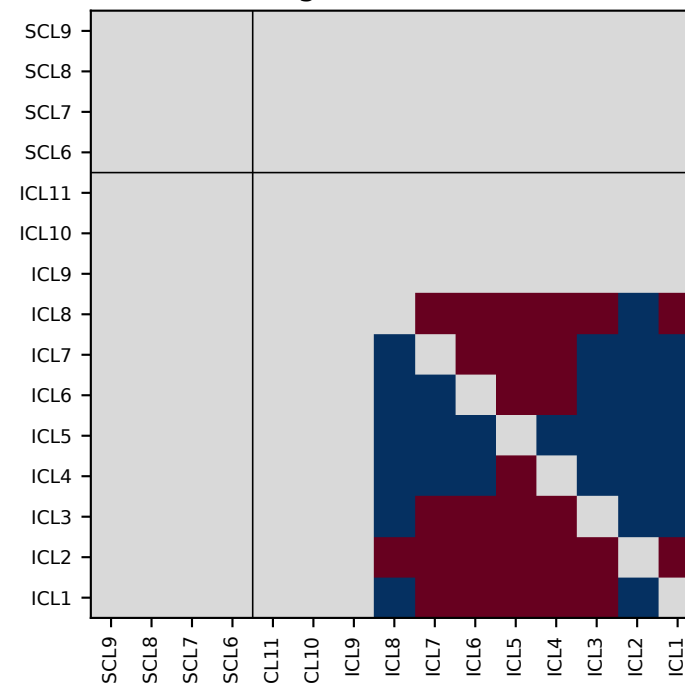
Mean normalized rank



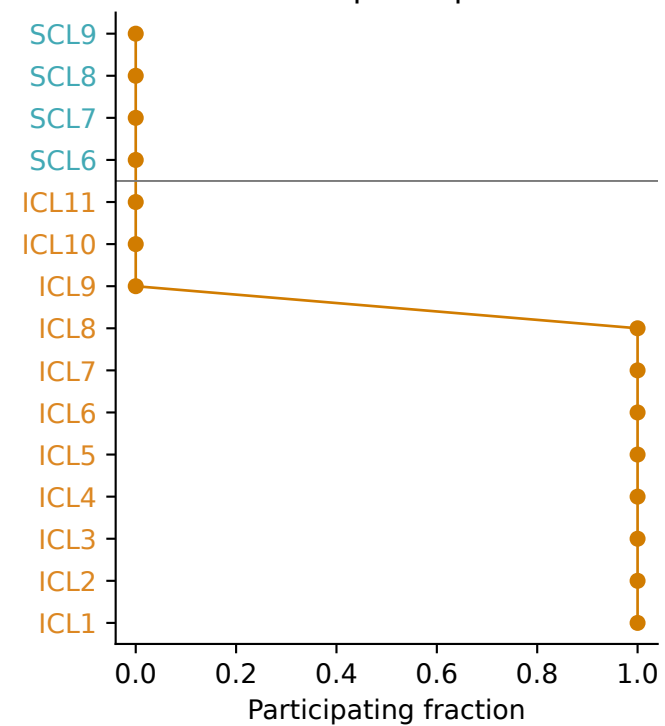
B-leading: within-shaft order



A-leading: within-shaft order



Contact participation



Fraction: row contact precedes column contact

● B-leading: 0 eligible events ● A-leading: 8 eligible events ■ Pair not estimable / different shafts